

# Deliverable 1 – Project Requirements

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**Course:** CIS106

**Assignment:** Deliverable 1 – Project Requirements

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## 1. What is a Web Server?

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A web server is a software application that delivers web pages and files to users through a web browser. When a user visits a website, the web server receives the request and sends the requested webpage back to the user.

Examples of web server software include Apache, Nginx, and Microsoft IIS.

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## 2. Different Web Server Applications

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### Apache HTTP Server

#### Definition

Apache is a free and open-source web server software used to host websites and web applications.

#### Project Website

<https://httpd.apache.org/>

#### Operating Systems

- Linux
- Windows
- macOS
- Unix

#### Latest Version

Apache HTTP Server 2.4

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### Nginx

#### Definition

Nginx is a web server that is also used as a reverse proxy, load balancer, and HTTP cache.

#### Project Website

<https://nginx.org/>

## Operating Systems

- Linux
- Windows
- macOS
- Unix

## Latest Version

Nginx 1.28

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## Microsoft IIS

### Definition

Internet Information Services (IIS) is a web server software created by Microsoft for Windows systems.

### Project Website

<https://www.iis.net/>

## Operating Systems

- Windows Server
- Windows

## Latest Version

IIS 10

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## 3. What is Virtualization?

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Virtualization is the process of creating virtual versions of computers, operating systems, storage devices, or networks. It allows multiple operating systems to run on the same physical computer.

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## 4. What is VirtualBox?

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VirtualBox is a free and open-source virtualization software developed by Oracle. It allows users to create and run virtual machines on a computer.

### Website

<https://www.virtualbox.org/>

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## 5. What is a Virtual Machine?

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A virtual machine (VM) is a software-based computer that runs inside another computer. It behaves like a real computer with its own operating system, storage, memory, and applications.

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## 6. Host Machine vs Guest Machine

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### Host Machine

The host machine is the physical computer that runs the virtualization software.

### Guest Machine

The guest machine is the virtual machine running inside the host machine.

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## 7. What is Debian?

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Debian is a free and open-source Linux operating system known for being stable, secure, and reliable. Many Linux distributions are based on Debian.

### Website

<https://www.debian.org/>

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## 8. What is a Firewall?

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A firewall is a security system that monitors and controls incoming and outgoing network traffic. It helps protect computers and networks from unauthorized access.

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## 9. What is SSH?

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SSH stands for Secure Shell. It is a network protocol used to securely connect to another computer over a network.

SSH is commonly used to remotely manage Linux servers.

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## 10. What is an IP Address?

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An IP address is a unique number assigned to a device connected to a network. It allows devices to communicate with each other over the internet or a local network.

Example:

```
192.168.1.1
```

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## 11. What is a Network Mask?

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A network mask, also called a subnet mask, is used to divide an IP address into network and host portions. It helps devices determine whether another device is on the same network.

Example:

```
255.255.255.0
```

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## 12. What is a Port?

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In networking, a port is a communication endpoint used by applications and services to send and receive data.

Examples:

- Port 80 → HTTP
- Port 443 → HTTPS
- Port 22 → SSH

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## 13. What is Port Forwarding?

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Port forwarding is a networking technique that redirects communication requests from one address and port number to another. It is commonly used to allow external access to services inside a private network.

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## 14. What is Localhost?

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Localhost is a hostname that refers to the current computer being used. It is commonly used for testing network applications locally.

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## 15. What does the IP Address 127.0.0.1 represent?

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The IP address **127.0.0.1** is called the loopback address. It points back to the same computer and is used to test networking services locally.

Example:

```
ping 127.0.0.1
```

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## 16. What is Git?

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Git is a distributed version control system used to track changes in files and projects. It helps developers manage source code and collaborate with others.

Website

<https://git-scm.com/>

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## 17. What is GitHub?

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GitHub is a cloud-based platform used to host Git repositories. It allows developers to store projects online, collaborate, and track changes.

Website

<https://github.com/>

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## 18. Concepts Researched After Reading the Tutorial

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### Reverse Proxy

A reverse proxy is a server that receives requests from users and forwards them to another server.

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### Load Balancer

A load balancer distributes network traffic across multiple servers to improve performance and reliability.

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### Open Source

Open source software is software whose source code is publicly available so anyone can study, modify, and distribute it.

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## Linux Distribution

A Linux distribution is a version of Linux that includes the Linux kernel along with software packages and tools.

Examples:

- Debian
- Ubuntu
- Fedora

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## Conclusion

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In this deliverable, I learned about web servers, virtualization, networking concepts, Linux systems, Git, and GitHub. These technologies are important for understanding how servers and networks work in real-world environments.